Lesson 3.2: Hallucinations: fabricated citations, numbers, quotes, and why confidence is not evidence

Claude Certified Associate – Foundations (CCAO-F) · Module 3: Evaluating & Validating Output · Lesson 2 · 15:55

Description

Claude can write a fluent, confident answer that contains details nobody ever wrote: a page number, a percentage, a quote. This lesson explains why. Claude writes its answer a piece at a time, so a sentence flows just as smoothly whether the fact inside it is real or not. The risk concentrates in specifics, and it climbs when a topic is newer than Claude's knowledge cutoff. Anthropic's own research shows that Claude's default is to say it can't answer, and how that default can misfire. You see where each idea comes from: the official exam guide, Claude Academy, the Claude Help Center, Anthropic's interpretability research and the Claude Platform Docs. Then the same three questions run twice in Claude Desktop, live. The first time, Claude answers from memory, and every specific it gives is checked against the real source with Ctrl+F. The second time, the source is attached and the request is built so a gap gets admitted instead of filled. The lesson ends with one exam-style scenario, reasoned out loud, and the cue to this lesson's quick check.

Learning outcomes

By the end of this lesson you can:

1. Explain what a hallucination is, and why fluent text can still be invented.
2. Spot the risky specifics (citations, numbers, quotes) and check each one against the source.
3. Set up a request so a gap gets admitted, not filled.

Chapters

Time	Chapter
00:00	Preview and outcomes
00:43	Why it matters
01:50	The idea: why fluent text can be invented
03:56	Straight from the source
05:39	Hands-on 1: three specifics, from memory
10:28	Hands-on 2: the same questions, with the source
14:07	Exam angle
15:08	Recap and quick check

The idea

What a hallucination is. Claude states something that isn't true, or isn't supported by its source, as if it were true. The details look real. They just aren't.

Why fluent text can be invented. Claude writes its answer a piece at a time, each time picking what tends to come next. Claude Academy puts it this way: it's closer to a vastly sophisticated autocomplete than to a search engine. Without search or a file, nothing gets looked up; it gets written. That one property gives you both the fluency and the hallucination, so nothing in the writing tells you whether a fact is real.

Where to look hardest: the specifics. Names, dates, numbers, citations, web addresses and quotes. That's where fabrication concentrates, and the more precise a claim, the more it needs checking. The trap: precise details are exactly what make an answer feel trustworthy.

When the risk climbs. When a topic is thin in what Claude learned: niche, local, or newer than its knowledge cutoff. Past that date, without search or a file, Claude has no information to go on, but it can still write about the topic fluently.

Why Claude often says "I can't answer", and why it sometimes doesn't. Anthropic's researchers looked inside the model and found that refusing to answer is Claude's default. Recognizing something familiar switches that default off. When Claude recognizes a name but knows nothing else about it, the switch can misfire, and Claude fills the gap with something plausible but untrue. So both outcomes are possible.

Confidence is not evidence. The confident tone is generated too. Anthropic wants Claude's confidence to match what it actually knows, and Claude tries, but a tone is not proof. Confidence is how an answer sounds. Evidence is something you can find in a source and check. A hedge isn't evidence either: "I believe" can still be wrong, and a flat statement can still be right. A hedge tells you to check; it doesn't do the checking.

The check (from Lesson 3.1). Find each specific in the source, for example with Ctrl+F. Every specific gets one verdict: it matches, it changed, or it's not in the source. Only the source can settle it.

The request that makes a gap visible. Three lines from Anthropic's *Reduce hallucinations* page, written for developers but just as useful in a chat:

Line in the prompt	The technique on the Docs page
Use only the attached guide.	External knowledge restriction
For each answer, quote the guide word for word and give the page.	Use direct quotes; verify with citations
If you can't find a supporting quote, write "Not in the guide" instead of answering.	Allow Claude to say "I don't know"

One honest limit. The same page says these techniques significantly reduce hallucinations, but they don't eliminate them. So you still check.

Straight from the source (what the lesson shows)

- **Exam guide, page 4:** "Exam items are written against these objectives." Domain 2, Output Evaluation and Validation (21% of scored items), second bullet: "Identify hallucinations, inconsistencies, and biases in responses". This course labels it task 2.2. The guide lists its task statements as unnumbered bullets, so 2.2 is our label, not the guide's.
- **Exam guide, page 7 (answer key for Sample 1):** "Language models can fabricate specific-looking details such as citation numbers, a hallucination." And: "Self-reported confidence (A, C) is not a reliable accuracy signal".
- **Claude Academy, AI capabilities and limitations, *Next Token Prediction*:** "Fabrication concentrates in specificity: names, dates, statistics, citations, URLs, quotes. The more precise a claim, the more it warrants verification." Its exercise, Probe 2 "Specificity under pressure": ask for checkable specifics and verify every one. The hands-on is this course's run of that probe, on material that can be checked on screen.
- **Claude Help Center, *Claude is providing incorrect or misleading responses*:** "Claude can display quotes that may look authoritative or sound convincing, but are not grounded in fact."
- **Anthropic research, *Tracing the thoughts of a large language model*, section "Hallucinations":** "in Claude, refusal to answer is the default behavior". Misfires "can occur when Claude recognizes a name but doesn't know anything else about that person", and then the model "proceeds to confabulate: to generate a plausible—but unfortunately untrue—response."
- **Claude Platform Docs, *Reduce hallucinations*:** allow Claude to say "I don't know" ("This simple technique can drastically reduce false information"), use direct quotes for factual grounding, verify with citations ("If it can't find a quote, it must retract the claim"), external knowledge restriction, and the note: "while these techniques significantly reduce hallucinations, they don't eliminate them entirely."

Key terms

- **Hallucination:** when Claude states something that isn't true, or isn't supported by its source, as if it were.
- **Specifics:** names, dates, numbers, citations, web addresses and quotes: where fabrication concentrates, so where you check first.
- **Knowledge cutoff:** the date Claude's training data ends. Without search or a file, it can't know anything after it.
- **Evidence:** something you can find in a source and check. Tone, hedges and self-ratings are not evidence.
- **Fabricated quote:** words in quotation marks that the source never said, including a paraphrase dressed as a quote.

What happened in the hands-on

Claude Desktop on Linux, Chat view, Sonnet 5.5 at Medium, recorded on 2026-10-06 on a real account. The job is real: the study notes for this lesson need three specifics, and every one can be checked against a real source. Nothing was planted, and nothing was re-run to get a different outcome. Both generations play in real time; only still pauses where nothing happened on screen were trimmed.

The prompt (the same three questions in both runs):

> I'm writing study notes for a lesson on hallucinations in my CCAO-F exam course. Please give me these three things exactly:
>
> 1. The exact wording of the CCAO-F exam guide's task statement about hallucinations, and the page it's on.
> 2. A word-for-word quote from Anthropic's "Reduce hallucinations" docs page about using direct quotes, with the page's URL.
> 3. Domain 2's weight on the exam, as a percentage.

Why these three are risky: item 1 is a quote plus a citation from a guide that came out in July 2026, after Sonnet 5.5's training data ends (June 2026). Item 2 is a quote plus a web address. Item 3 is a plain number.

Run 1: from memory

Setup. Nothing needed keeping, so this ran in an incognito chat (the ghost icon). In the + menu, Web search was ticked, and it was unticked; the menu was reopened to make sure. No file was attached, and the connectors to other apps were switched off before recording. So the answer could only come from Claude's training.

1. **The reply declined two of the three.** Its first line: it would rather say what it can't verify than hand over something plausible-sounding. Item 1: it doesn't have the guide, and "Anything I wrote would be invented." Item 3: "I won't guess a percentage." That's the trained default from the research, working as intended.
2. **Item 2 came from memory, with a warning.** Claude gave an address and a quote, said it "may hallucinate citations", and asked for a comparison against the live page. A grey "Used a tool · Failed" row in the reply was an app hiccup: nothing was searched, the Web search card said search was off, and the answer cites no sources.

3. **Check, item 1.** Claude gave no wording, so there was nothing of its own to trace. In the guide, Ctrl+F on the start of the real line: 1 match, page 4, the second bullet of Domain 2.
4. **Check, item 2, the address.** The *Reduce hallucinations* page lives at platform.claude.com today. Claude's address, typed into the address bar: `docs.claude.com/en/docs/test-and-evaluate/strengthen-guardrails/reduce-hallucinations`. It redirects to the same page. An old address that still works: **Matches.**
5. **Check, item 2, the quote.** Ctrl+F on the Docs page: "extract word-for-word quotes first", 1 match in the direct-quotes bullet. The whole first sentence, as it came back: 1 match. Claude wrote "20K" with a capital K; the page uses "20k", and the browser's find ignores capitals. Every word is the same. The second sentence, "This grounds its responses in the actual text, reducing hallucinations.": 1 match. **Matches.**
6. **Check, item 3.** Guide page 4: the Domain 2 heading says 21%. Claude gave no number, so there was nothing of its own to check.
7. **The tally.** Two specifics declined: the trained default, holding. One given from memory, with a warning, and it happened to be right; we only know that because it was checked. Nothing was invented. But a default isn't a guarantee: the research shows how it can misfire. So don't build your work on hoping it holds; build the request so a gap can't be filled.

Run 2: with the source

Setup. The first chat was closed and a fresh incognito chat opened, so the first answer couldn't steer this one. Same model, Web search still off. The guide itself was attached with **Add files or photos** (the PDF). The prompt: the same three questions, pasted word for word, plus three new lines from the Docs page:

> Use only the attached guide. For each answer, quote the guide word for word and give the page. If you can't find a supporting quote, write "Not in the guide" instead of answering.

Item 2 is the honest test: its answer isn't in the guide at all. While generating, the status line said Claude was locating exact quotes and page references within the guide.

8. **The reply.** Item 1: "Identify hallucinations, inconsistencies, and biases in responses", a bullet under "Domain 2: Output Evaluation and Validation (21%)", on page 4. Item 2: "Not in the guide." Item 3: 21%, with two quotes, the blueprint table row and the domain heading, both on page 4. Claude added a note of its own: the page numbers are its count from the footers, so check page 4 against your PDF.
9. **Check, item 1.** Ctrl+F on the whole quote, exactly as it came back: 1 of 1, page 4. **Matches.**
10. **Check, item 3.** The domain heading: 1 of 1, page 4. The table row, typed exactly as it appears in the answer: 1 of 1, page 4. **Both match.** Checking took seconds, because a quote is easy to find.
11. **Item 2, "Not in the guide."** The most valuable line in the answer. The quote asked for lives on a different page, and instead of filling the gap, Claude said so, because the prompt allowed it. If you need that quote, give Claude the Docs page, or let it search.

Side by side. Same model, same app, same three questions. From memory: two honest refusals, and one remembered quote that turned out right, with nothing to point to. With the source: every specific with its quote and page, and an honest gap where the source is silent. The model didn't get smarter. The request got checkable.

The exam link

Domain 2, Output Evaluation and Validation (21% of scored items). This lesson serves the bullet this course labels **2.2**: "Identify hallucinations, inconsistencies, and biases in responses". The guide lists its task statements as unnumbered bullets; the number 2.2 is this course's label, not the guide's. This lesson takes the first word, hallucinations. Inconsistencies and bias come in Lesson 3.3, and the full fact-checking toolkit in Lesson 3.4.

The lesson's practice scenario (original, written for this course, not from the exam). A communications lead asks Claude for a short quote from the company CEO's keynote, for the all-staff newsletter. She doesn't attach the transcript. Claude returns a vivid line in quotation marks. What's the most appropriate next step? A: use it, because it sounds just like the CEO. B: ask Claude whether the quote is accurate. C: check the line word for word against the keynote transcript before using it. D: keep it, but add "paraphrased" after it. The answer is **C**. Start with what kind of specific this is: a quote, with no source given, the riskiest kind. A trusts the sound of it, and sounding right is exactly what generated text does. B asks the writer to grade itself, and self-reported confidence isn't evidence. D is a trap: a made-up line with a label is still made up. C is the only one that looks for evidence.

Quick check

Before you move on, take this lesson's quick check: a few short questions, a couple of minutes. The exam-style question in this lesson, the quick checks, the module quizzes and the practice exams are all original, written for this course against the official blueprint. They are never copies of the real exam.

Sources (all checked 2026-10-06)

- **Anthropic, *Claude Certified Associate – Foundations Exam Guide*.** Version 1.0, effective July 2026. §6 Domain 2 and its detailed objectives (p. 4), §8 Sample 1 (p. 6) and its answer key and rationale (p. 7). Official download: https://everpath-course-content.s3-accelerate.amazonaws.com/instructor/6nizmqk8tpzpfjvt6qmmav7rh/public/1783542847/Claude+Certified+Associate+%E2%80%93+Foun
- **Claude Academy, AI capabilities and limitations,** *Next Token Prediction* (Summary, Key takeaways, The Verification Test, Probe 2): https://academy.claude.com/courses/ai-capabilities-and-limitations/next-token-prediction
- **Claude Help Center,** *Claude is providing incorrect or misleading responses. What's going on?* (updated March 16, 2026): https://support.claude.com/en/articles/8525154-claude-is-providing-incorrect-or-misleading-responses-what-s-going-on

- **Anthropic research,** *Tracing the thoughts of a large language model* (March 27, 2025), section "Hallucinations": https://www.anthropic.com/research/tracing-thoughts-language-model
- **Claude Platform Docs,** *Reduce hallucinations* (the old docs.anthropic.com and docs.claude.com addresses redirect here): https://platform.claude.com/docs/en/test-and-evaluate/strengthen-guardrails/reduce-hallucinations
- **Background, not shown in the lesson:**
  - Claude Platform Docs, *Models overview* (Sonnet 5.5: reliable knowledge cutoff and training data cutoff June 2026): https://platform.claude.com/docs/en/about-claude/models/overview
  - Claude Academy, AI capabilities and limitations, *Knowledge*: https://academy.claude.com/courses/ai-capabilities-and-limitations/knowledge
  - Claude Academy, AI capabilities and limitations, *When Properties Collide*: https://academy.claude.com/courses/ai-capabilities-and-limitations/when-properties-collide
  - Anthropic, *Claude's constitution* (honesty: calibrated): https://www.anthropic.com/constitution
  - Anthropic's official CCAO-F prep course, Course 3, *Evaluating & Validating Claude's Output* (public course page): https://anthropic-partners.skilljar.com/evaluating-validating-claudes-output
  - Claude Academy, AI Fluency: Framework and foundations, *Capabilities and limitations*: https://academy.claude.com/courses/ai-fluency-framework-foundations/capabilities-limitations
- **The app itself:** Claude Desktop on Linux, Chat view, Sonnet 5.5 Medium, observed 2026-10-06: incognito chats, the + menu (Add files or photos, Web search), the file chip, the status lines, the copy buttons and the "Used a tool · Failed" row. Menus, labels and status lines can change between versions.

"Matches / Changed / Not in source" and the three-line request are this course's teaching tools, built from the sources above (the three lines map one to one to techniques on Anthropic's *Reduce hallucinations* page). They are not an Anthropic checklist. The hands-on material is real: the official exam guide, Anthropic's own Docs page, and a real job for this course. Every verdict is the course author's, recorded during the take.

What you see in the lesson is a real account. Both runs used incognito chats, so they left no chat history and created no artifact. Settings, Account and Memory were never opened. The connectors were switched off before recording and switched back on afterwards. The account's suggestion chips on the new-chat screen are covered in the edit.